Cohort A: Normal CXR
(Stochastic Invariance Transforms)

Original Normal CXR



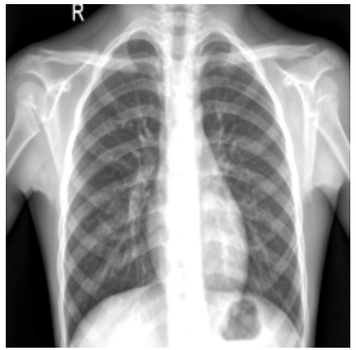
Random Affine (±10°, Scale)



Horizontal Flip (p=0.5)



Contrast & Brightness Jitter



Random Resized Crop (0.8-1.0)

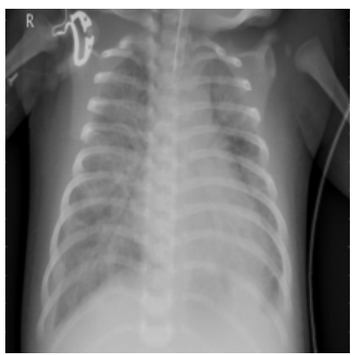


Gaussian Blur (Scanner Variance)

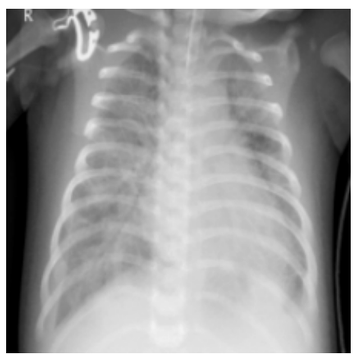


Cohort B: Pneumonia CXR
(SSL Multi-View / Multi-Crop)

Original Bacterial CXR
(Right Lobar Infiltration)



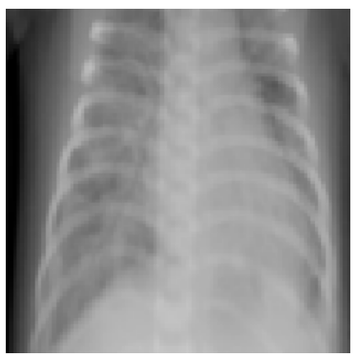
SSL Augmented View 1 (v1)
(Global Crop + Jitter)



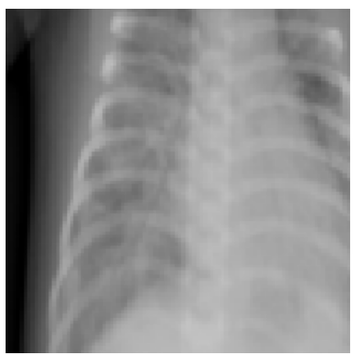
SSL Augmented View 2 (v2)
(Affine + Flip + Jitter)



DINO Local Crop 1 (96x96)
(Focal Consolidation Focus)



DINO Local Crop 2 (96x96)
(Perihilar Context)



Supervised Augmented Sample
(Affine + Flip + Jitter)

